nouPix

GETTING STARTED · V1.6.0

FROM ZERO TO LIGHT, STEP BY STEP.

Install nouPix, draw your fixtures, patch them, play content in zones and layers, store scenes, and run the show from a controller or a phone.

nouPix 1.6.0 · macOS · Windows · Linux · Raspberry Pi
Free and open source · GNU AGPL-3.0 · NOULED.STUDIO
noupix.nouled.com



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START HERE Steps 03 to 17 build one small project from an empty canvas: four bars, a matrix, a ring and a wave, in two zones. Every picture is the real app.

01

INSTALL AND OPEN

GET NOUPIX RUNNING

MACOS

macOS 12 or newer · Apple silicon or Intel

1. Download the **.dmg** for your Mac from the download page.
2. Open it and drag **nouPix** onto Applications.
3. First start: click **Done** on the warning, then **System Settings → Privacy & Security → Open Anyway**.

WINDOWS

Windows 10 or newer · 64-bit

1. Download the **installer .exe**.
2. If SmartScreen stops it: **More info → Run anyway**.
3. Open nouPix from the Start menu. Allow it through the firewall on **private networks**, or node discovery won't find your controllers.

LINUX · PI

Any computer · Python 3.8+ and Chrome or Chromium

1. Get the source and run it:

```
$ git clone https://github.com/nouled-studio/nouPix.git
$ cd nouPix
$ python3 noupix.py
```

2. nouPix opens in its own window. Keep the terminal open: closing it stops the output.

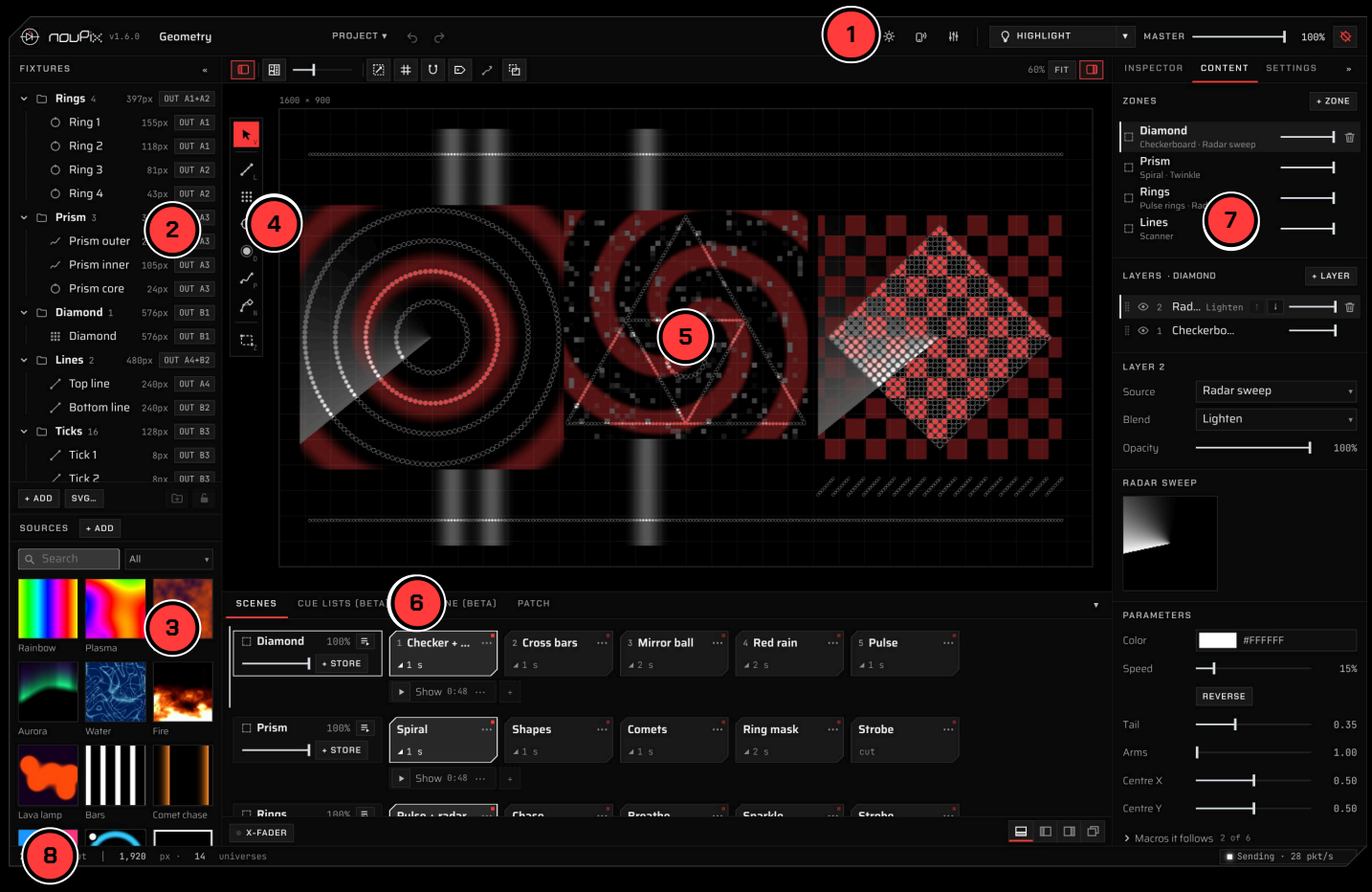
FIRST START nouPix opens with a demo layout, already playing. Look around, then start your own project (step 03).

02

THE WINDOW

EVERYTHING ON ONE SCREEN

1. **Top bar**: project, undo, black or white, remote, MIDI, highlight, master, blackout
2. **Fixtures**: everything on the stage, in groups
3. **Sources**: effects, your shaders, images, video, capture, streams
4. **Tools**: select, line, matrix, circle, dot, path, pen, zones
5. **Stage**: your layout, playing live
6. **Dock**: Scenes · Patch, with Discovery, the Channel monitor and Output settings
7. **Right panel**: Inspector · Content · Settings
8. **Status bar**: frame rate, pixels, universes, what goes out



TIP

Right-click anything for its options. H hides the right panel, Shift H the left, F fits the stage.

03

A NEW PROJECT

START FROM AN EMPTY CANVAS

1

Project ▾ → New empty project. It starts with one zone playing Rainbow, so every pixel you draw lights up.

2

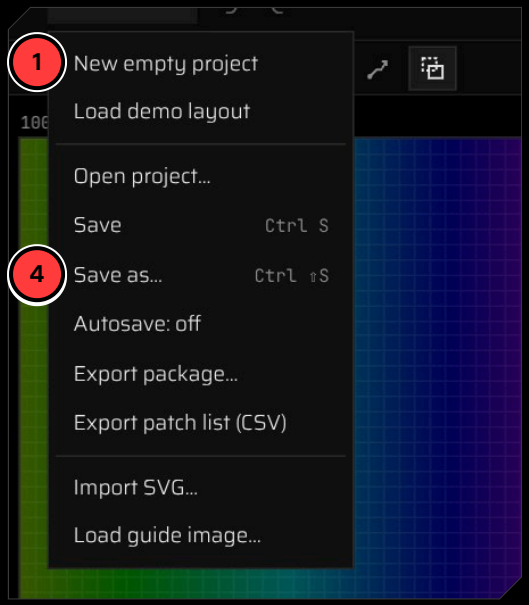
Name it: click the name in the top bar.

3

Settings → Canvas: the size of your canvas, in any unit you like (for example 1000 × 640). The frame rate is under Output.

4

Project ▾ → Save as... writes a .noupix file. Turn on Autosave and every change is saved to it.





TIP Ctrl S saves, Ctrl Z undoes. Projects open with Project ▾ → Open project...; the examples on GitHub are ready to open.

04

DRAW FIXTURES

DRAW WHAT YOU BUILT

1

Pick a tool: `L` line or strip, `M` matrix, `C` circle or ring, `D` dot, `P` path, `N` pen for curves.

2

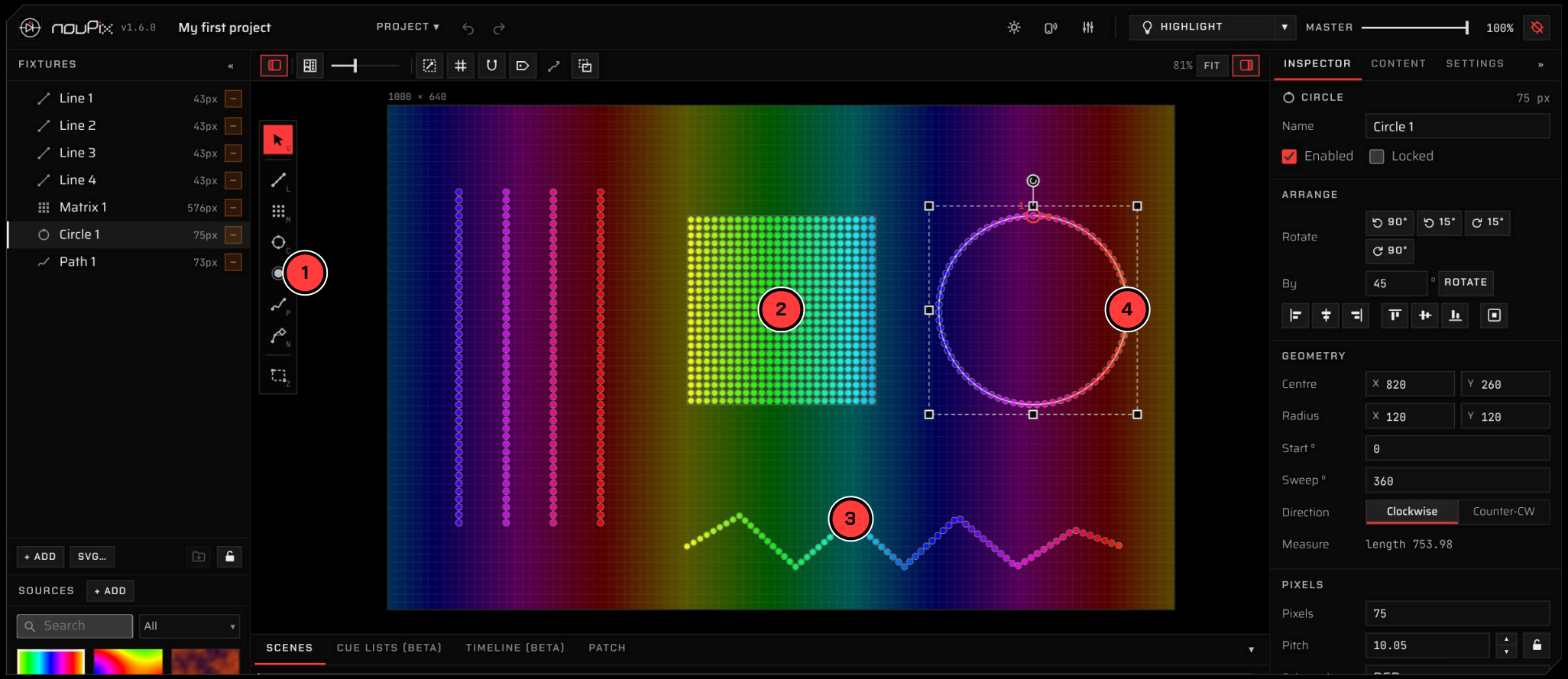
Drag on the stage: a line from end to end, a matrix as a rectangle, a circle from its centre outwards.

3

Path: click the points, then `Enter`. Pen: click for corners, drag for curves.

4

`V` selects and moves. `Shift` snaps to 45°, `Alt` turns snapping off. A click with a tool drops a default fixture.



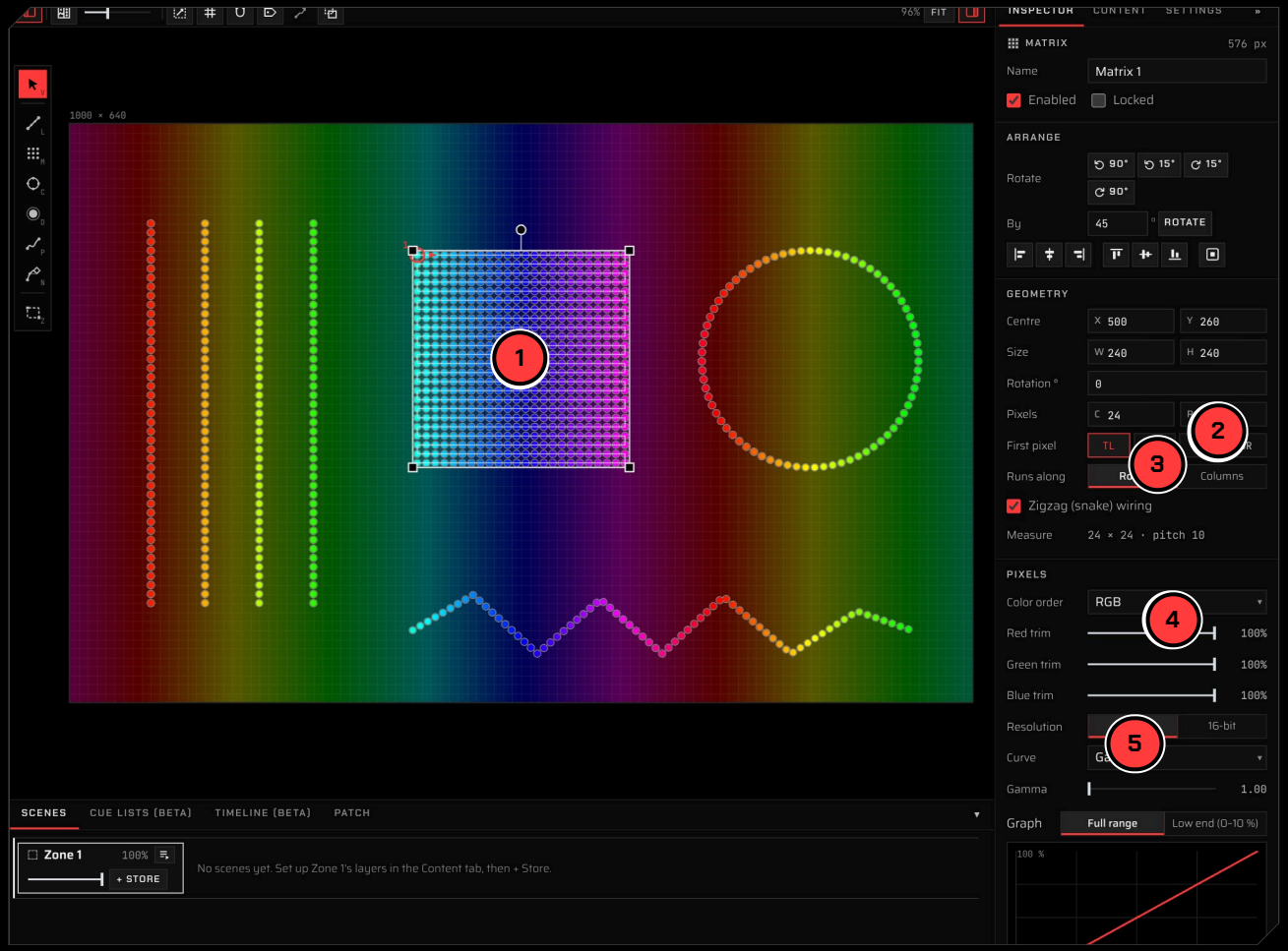
TIP Each new fixture appears in the Fixtures list on the left, with its pixel count. A Dot is one pixel as big as its light (a PAR, a wall washer): set its Size in the Inspector.

05

SET THEM UP

PIXELS, WIRING, COLOUR

1. **Select a fixture**: its settings open in the Inspector.
2. **Pixels**: the count, or columns × rows for a matrix.
3. **First pixel and wiring**: which corner, rows or columns, zigzag.
4. **Colour order** (RGB, GRB, RGBW…) and trims per colour.
5. **Resolution** 8 or 16-bit, and the dimming curve with its graph.
6. **Pitch lock** on a strip or ring: its length sets the pixels, like a strip cut to length.



LINE 1 · PITCH LOCKED



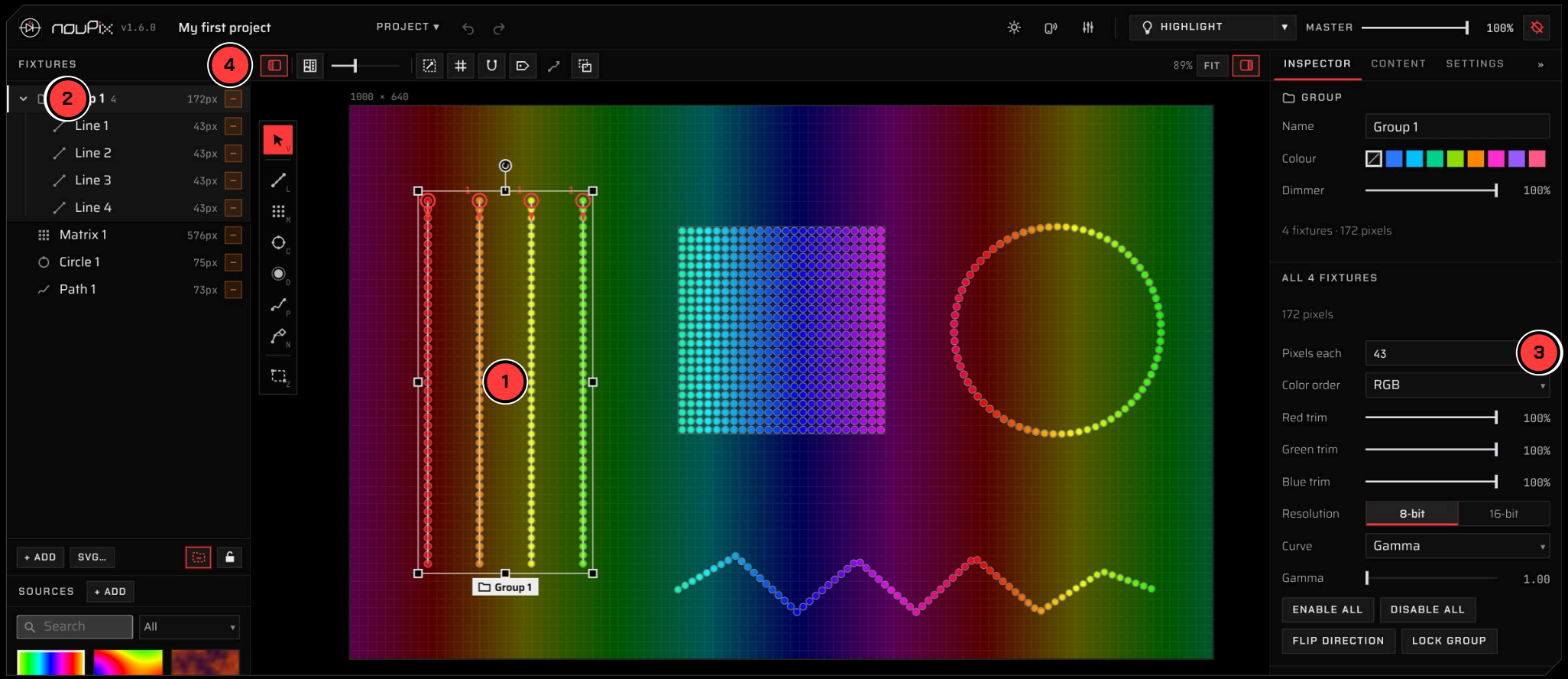
TIP Select several fixtures to set their pixels, colour order or gamma all at once.

06

ARRANGE AND GROUP

KEEP THE LAYOUT TIDY

1. Select several: `Shift` - click, or drag a box around them.
2. Group them with `Ctrl` `G` : the list folds them under one name, with its own dimmer.
3. Rotate, align and set pixels for all of them at once.
4. Lock with `Ctrl` `L` so nothing moves by accident.
5. Copy, paste and duplicate: `Ctrl` `C` `Ctrl` `V` `Ctrl` `D` .



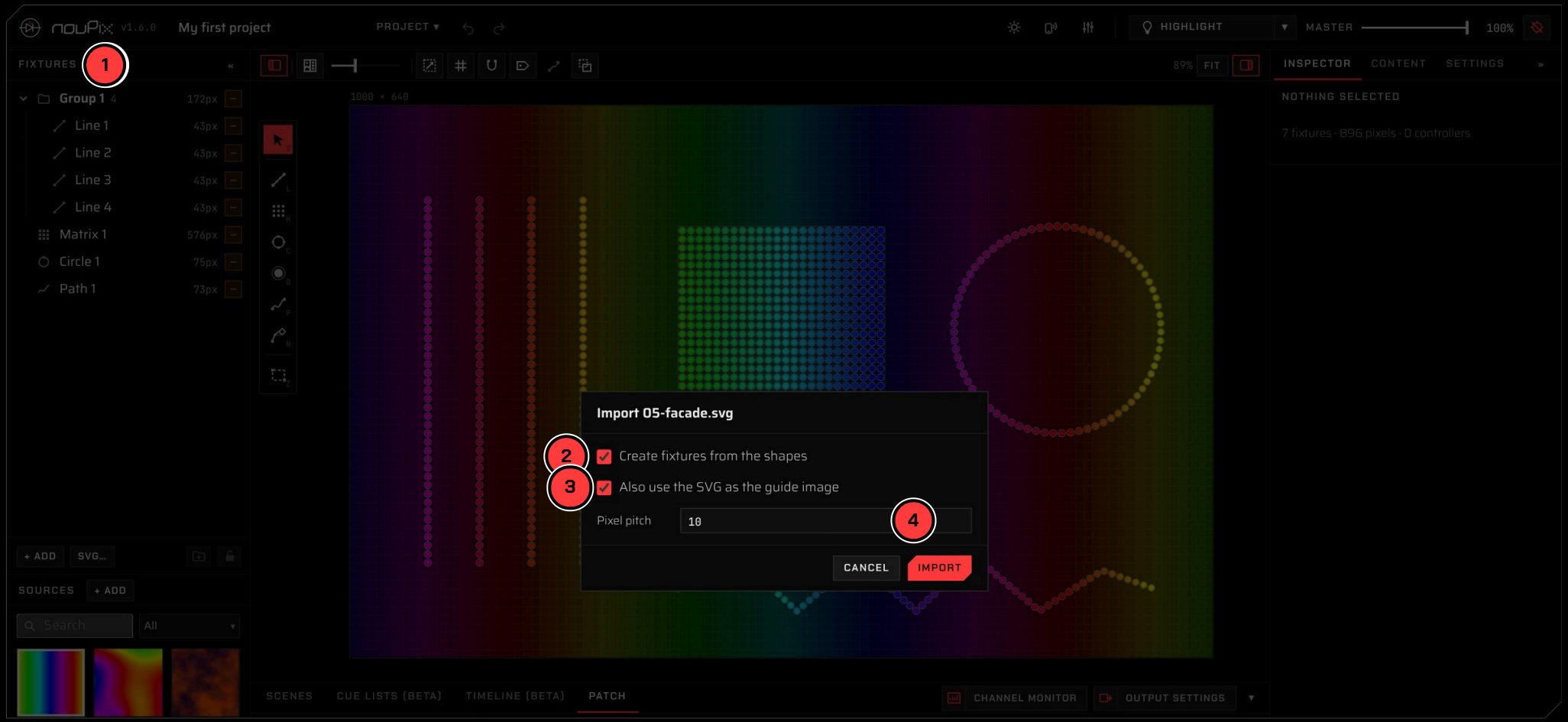
TIP Double-click a fixture inside a group to pick just that one. `R` turns the selection 90°.

07

IMPORT A DRAWING

START FROM YOUR SVG

1. SVG… at the top of the Fixtures list, or Project ▾ → Import SVG…
2. Create fixtures: every circle, arc, line and path of the drawing becomes one.
3. Use it as the guide image, under the fixtures, to check them against the building.
4. Pixel pitch: the spacing of the LEDs; it sets each fixture's pixel count.



TIP Export the façade from your drawing or 3D software as SVG, one stroke per LED strip.

08

FIND CONTROLLERS

DISCOVER THE DEVICES

1

Connect the controllers and this computer to the same network, with addresses in the same range (for example 10.0.0.x, or 2.x.x.x).

2

Patch tab → Discovery… → Discover: ArtPoll goes out on the networks ticked next to it.

3

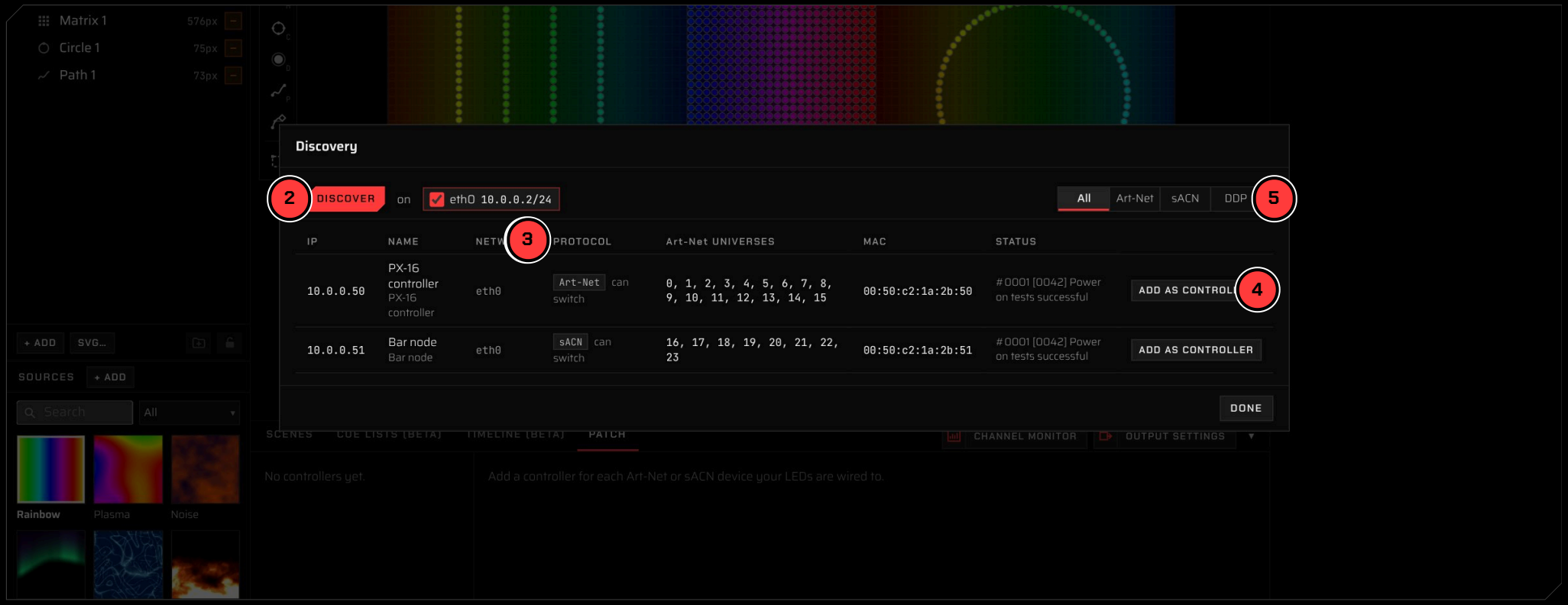
Every device that answers is listed with its IP, name, protocol (Art-Net, sACN or DDP) and universes. WLED too, found by its name.

4

Add as controller: it goes to the Patch list on its protocol, ready to patch; a WLED filled in completely.

5

All · Art-Net · sACN · DDP shows only the devices for one protocol.



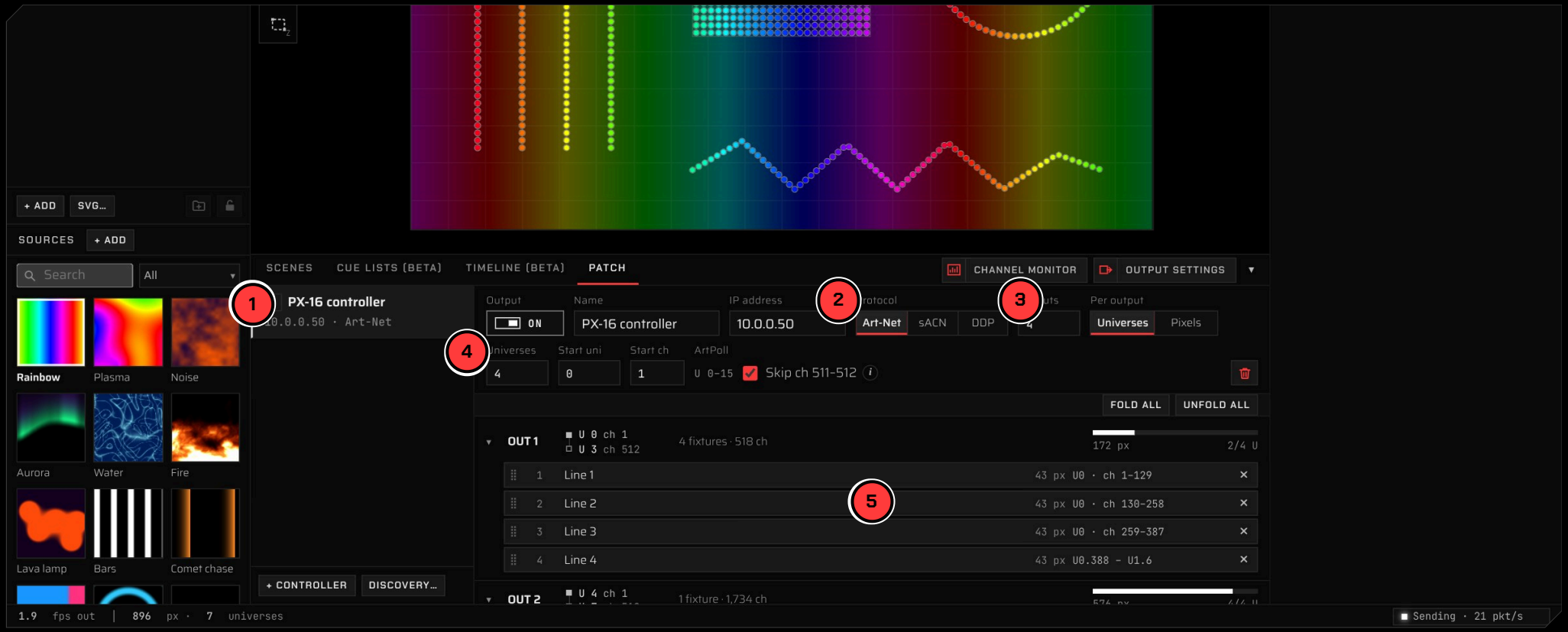
TIP Not found? Allow nouPix through the firewall on private networks, and give this computer an address in the device's range. + Controller adds one by hand.

09

PATCH

FIXTURES ONTO OUTPUTS

1. The **controller list**: name, IP and protocol. `Ctrl` or `Shift` selects several, `Del` removes them.
2. **Protocol** per controller: Art-Net, sACN and DDP go out together.
3. **Outputs** by **Universes**, or by **Pixels**: give the pixels per output and the pixel type, nouPix works out the universes.
4. **Start uni** and **Start ch**: where the controller begins. Each output shows where it starts and ends.
5. **Drag fixtures** from the list onto an output. The order on an output sets the addresses.



TIP Right-click a fixture → **Patch to…** The badge next to each fixture in the list shows its output.

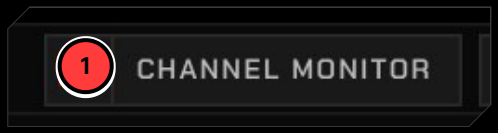
10

CHANNEL MONITOR

CHECK WHAT GOES OUT

1. **Channel monitor** at the right of the Patch tab: a window of its own, like the MIDI window.
2. **Pick a universe**: its 512 channels, live.
3. **Point at a channel** to see which fixture and pixel drive it.
4. The **status bar** shows how many packets go out each second.





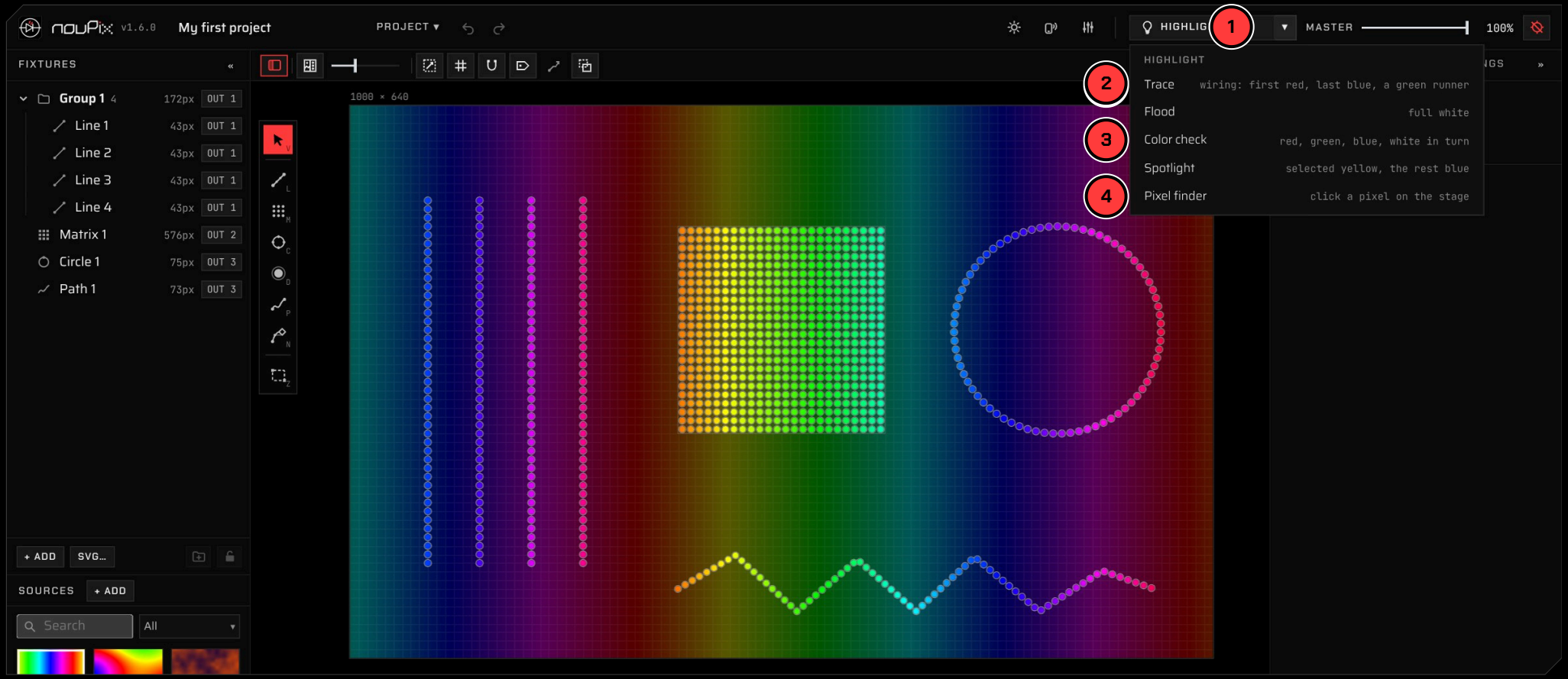
TIP Put it on a second screen. Patched fixtures light it even with no controller on the network: check the patch at your desk, before the site.

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HIGHLIGHT ON SITE

CHECK EVERY FIXTURE

1. **Highlight** in the top bar, or `I`: the content stops and test patterns light the fixtures.
2. **Trace**: first pixel red, last blue, a green runner along the wiring.
3. **Flood**, **Color check**, **Spotlight** (the selected ones yellow, the rest blue).
4. **Pixel finder**: click a pixel on the stage to light only that one.



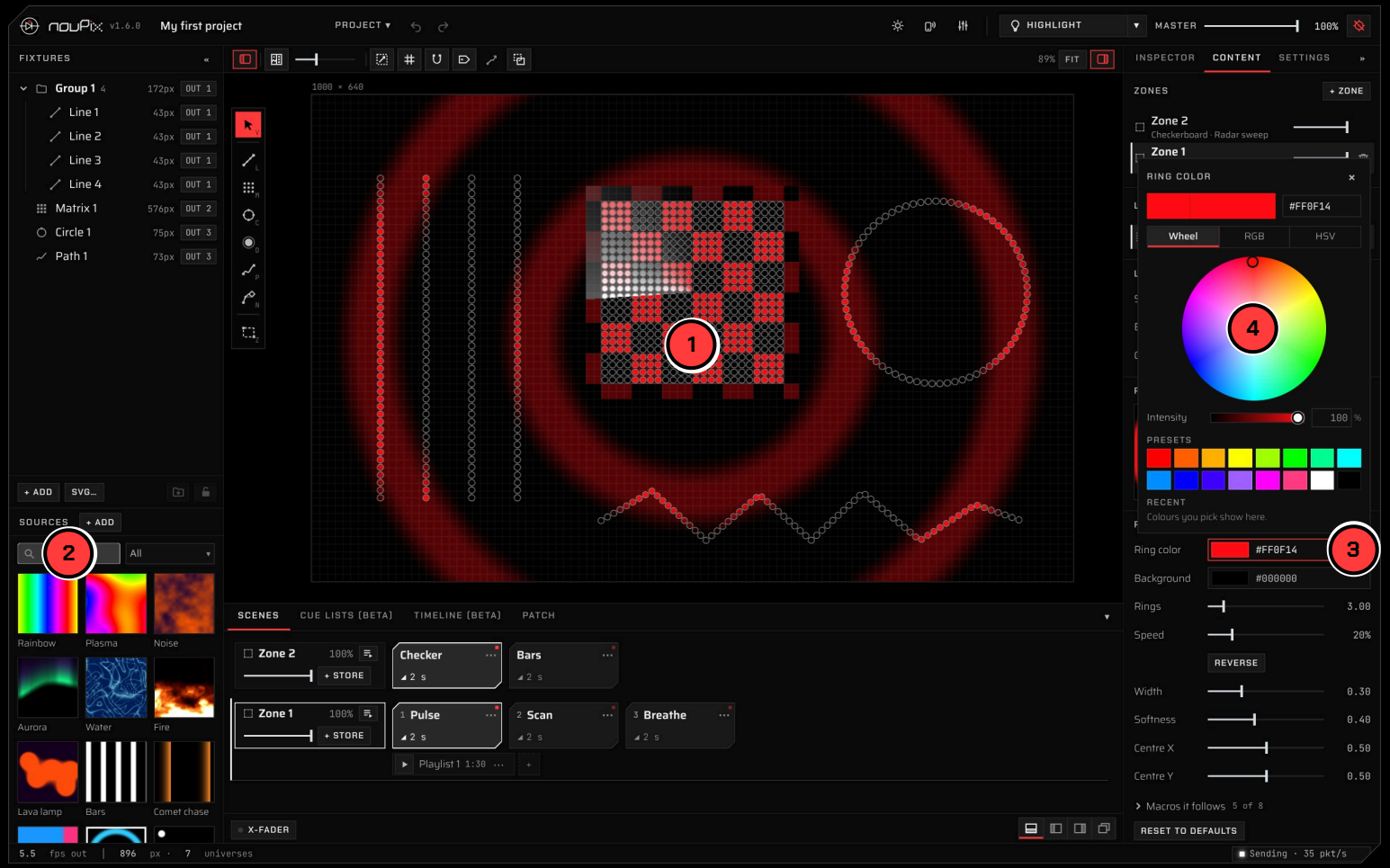
TIP Trace shows at once where a strip starts and whether it runs the right way.

12

PLAY CONTENT

CLICK A SOURCE, IT PLAYS

1. Click a source: it plays on the selected zone, and the stage shows it on the pixels.
2. Search, or show one use: Architectural, Ambient, Show, Utility.
3. Its controls in the Content tab. Double-click a value to type it.
4. Colours: wheel, RGB, HSV, presets and recent colours.



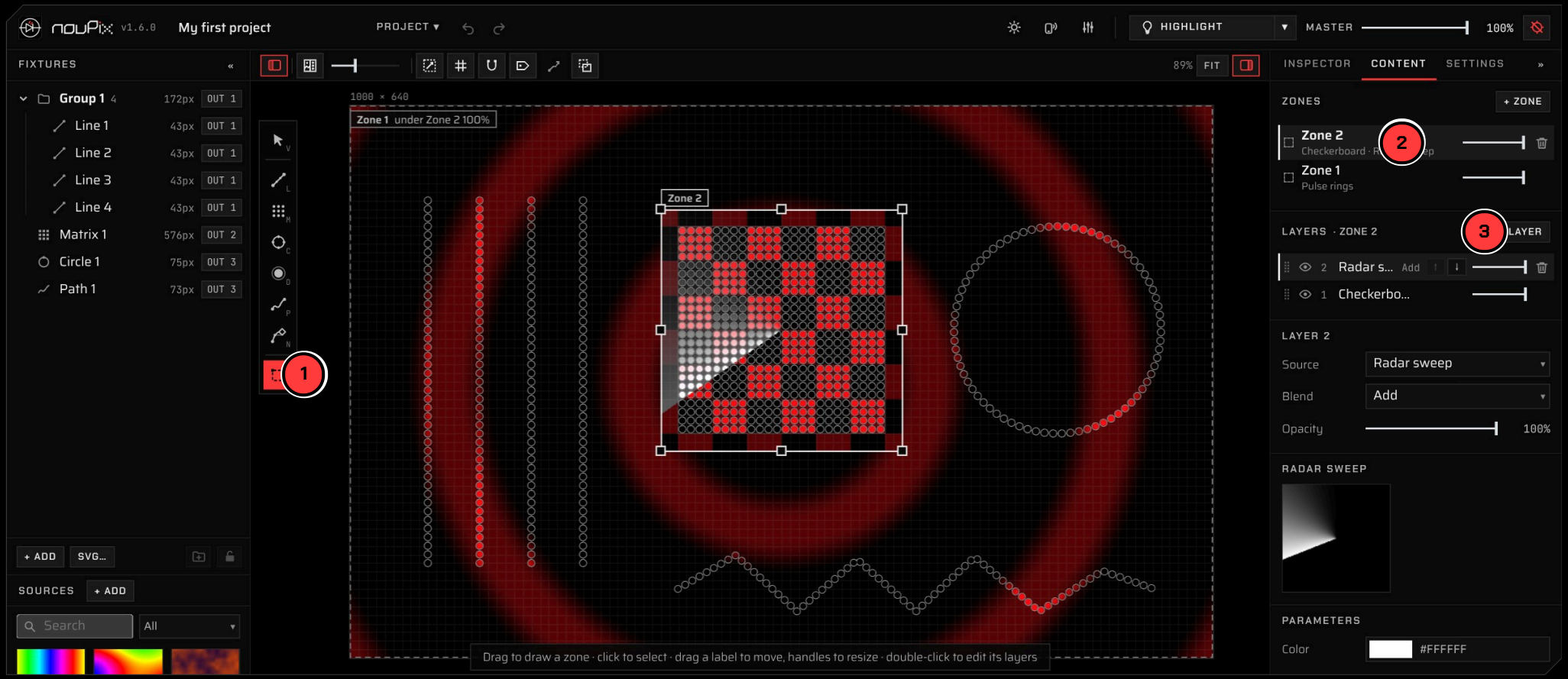
TIP `Shift` + scroll over a slider steps it; angles have buttons every 45°.

13

ZONES AND LAYERS

DIFFERENT CONTENT, DIFFERENT PARTS

1. **Z Zones tool**: drag a zone around part of the stage.
2. **Zones list**: the upper zone covers the lower. Drag to reorder.
3. **+ Layer** stacks another source in the zone.
4. **Blend** (Add, Screen, Multiply, Lighten…) and **opacity** per layer.



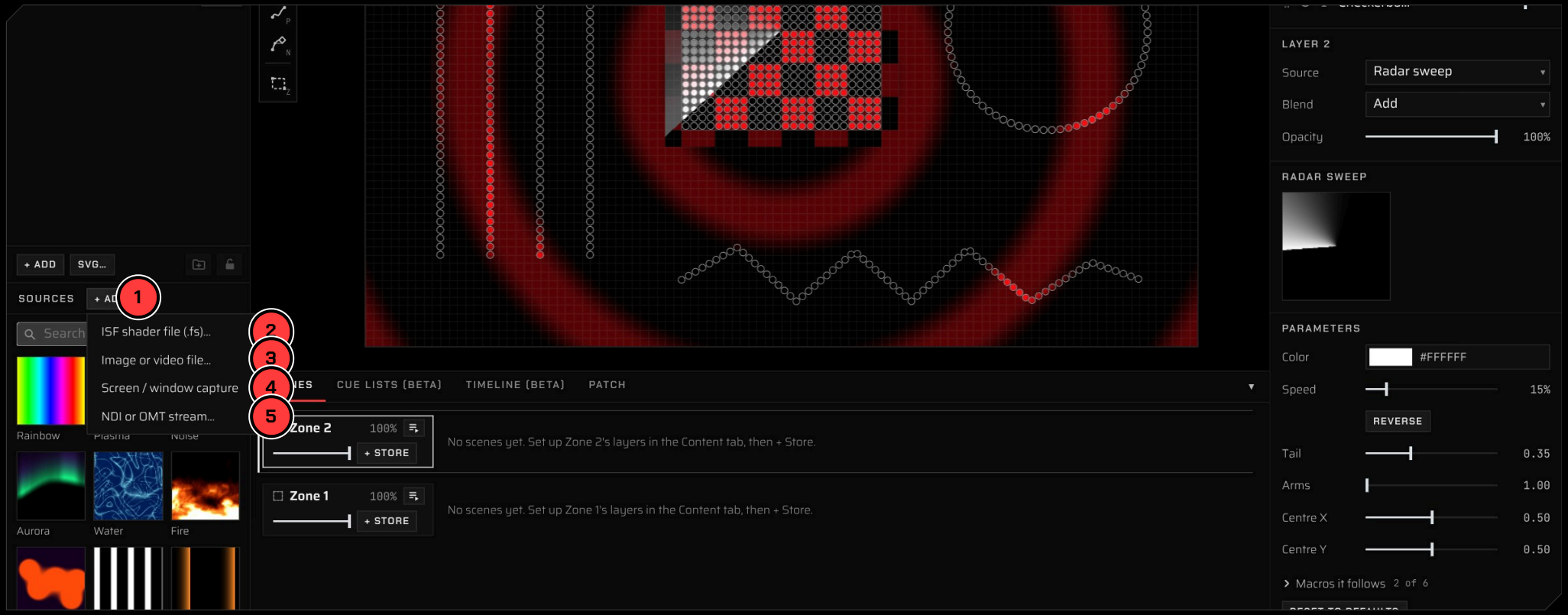
TIP Here Zone 2 plays a red checkerboard with a white radar sweep added on top, over the matrix only.

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YOUR OWN CONTENT

SHADERS, VIDEO, CAPTURE, STREAMS

1. + Add above the sources.
2. ISF shader file (.fs): your own generators, with their controls.
3. Image or video: play, pause, trim, loop or once, with or without sound.
4. Screen / window capture: for example the output of other VJ software.
5. NDI or OMT stream: from Resolume, TouchDesigner, OBS or another computer.



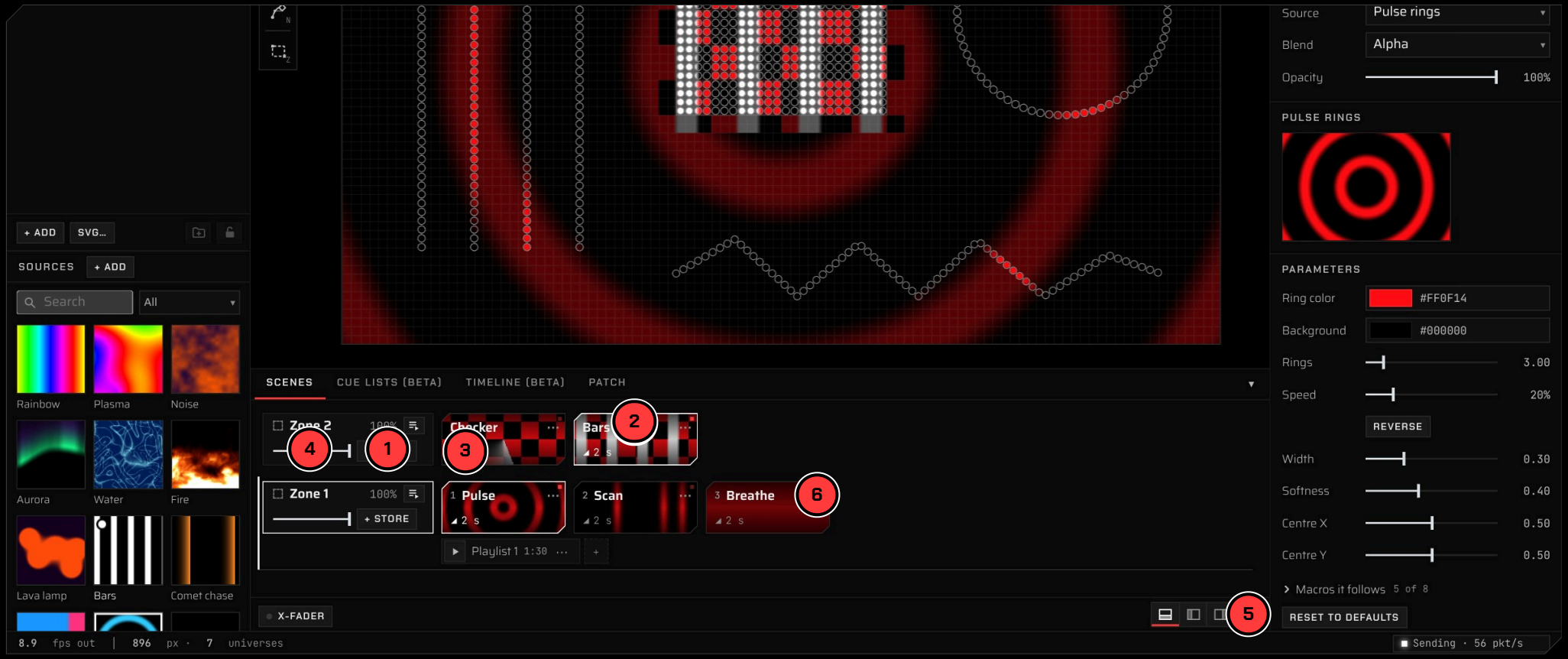
TIP Images and videos are copied into nouPix's media library, so projects open with them. NDI needs NDI Tools installed, OMT its library (see the user guide).

15

SCENES

STORE A LOOK, FADE BACK TO IT

1. + Store keeps the zone's layers as a scene.
2. Click a scene to fade to it; `Shift` - click cuts.
3. Drag the fade time on its button to change it.
4. Level of each zone, live.
5. Open the Scenes window for a second screen.
6. ⋯ → Take snapshot: shown on its button.



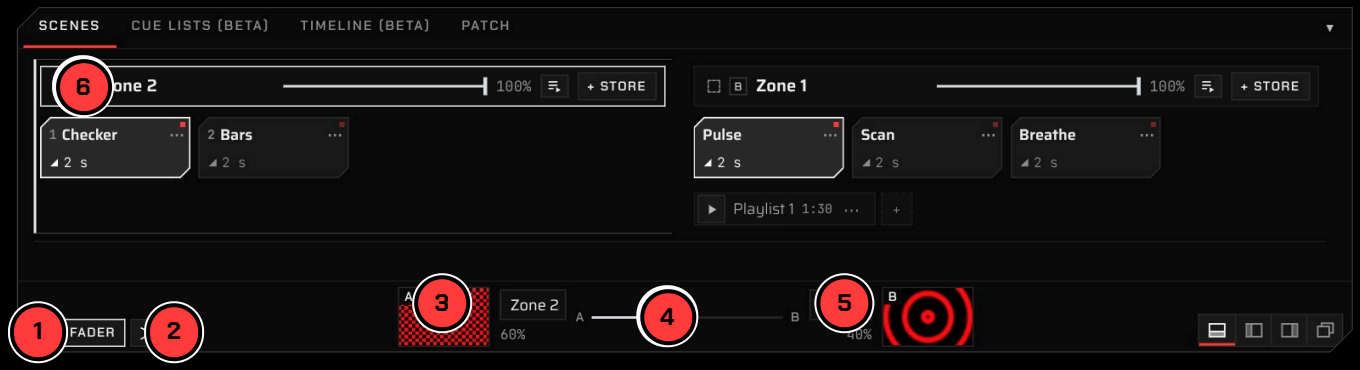
TIP Keys `1`-`9` play the scenes of the selected zone. Settings → Scenes → Auto snapshot takes one on every + Store.

16

X-FADER

ONE FADER BETWEEN TWO ZONES

1. **X-Fader**, bottom left of the Scenes panel, turns it on: A's and B's scenes side by side. Make both full size when asked.
2. **Mode**: Mix, Dip through black, or Cut at the middle.
3. **Live previews** of A and B, even the one faded out.
4. **The fader**: all A on the left, all B on the right.
5. **Choose the zones** for A and B from their lists.
6. **A and B**: each zone's own level is the most it reaches; the X-Fader mixes within it.







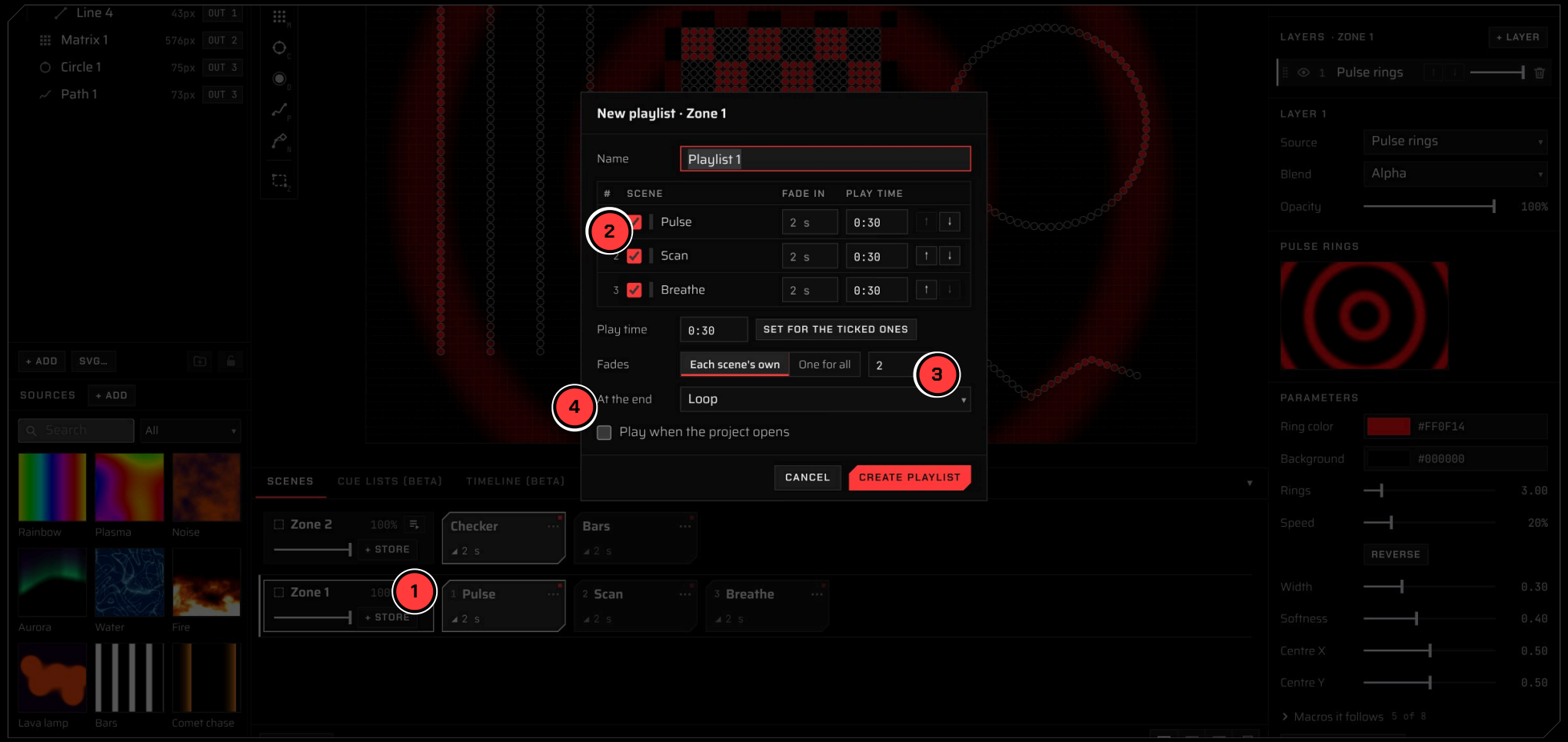
TIP Learn a MIDI fader onto it like any slider. Settings → Remote control → X-Fader on the Remote page puts it on the phones too.

17

PLAYLISTS

A SHOW THAT RUNS BY ITSELF

1. The playlist button next to + Store opens New playlist.
2. Pick the scenes, their play times and fades.
3. Play, pause, next. In a loop, once, or handing over to another playlist.
4. Tick Play when the project opens: the show starts by itself.



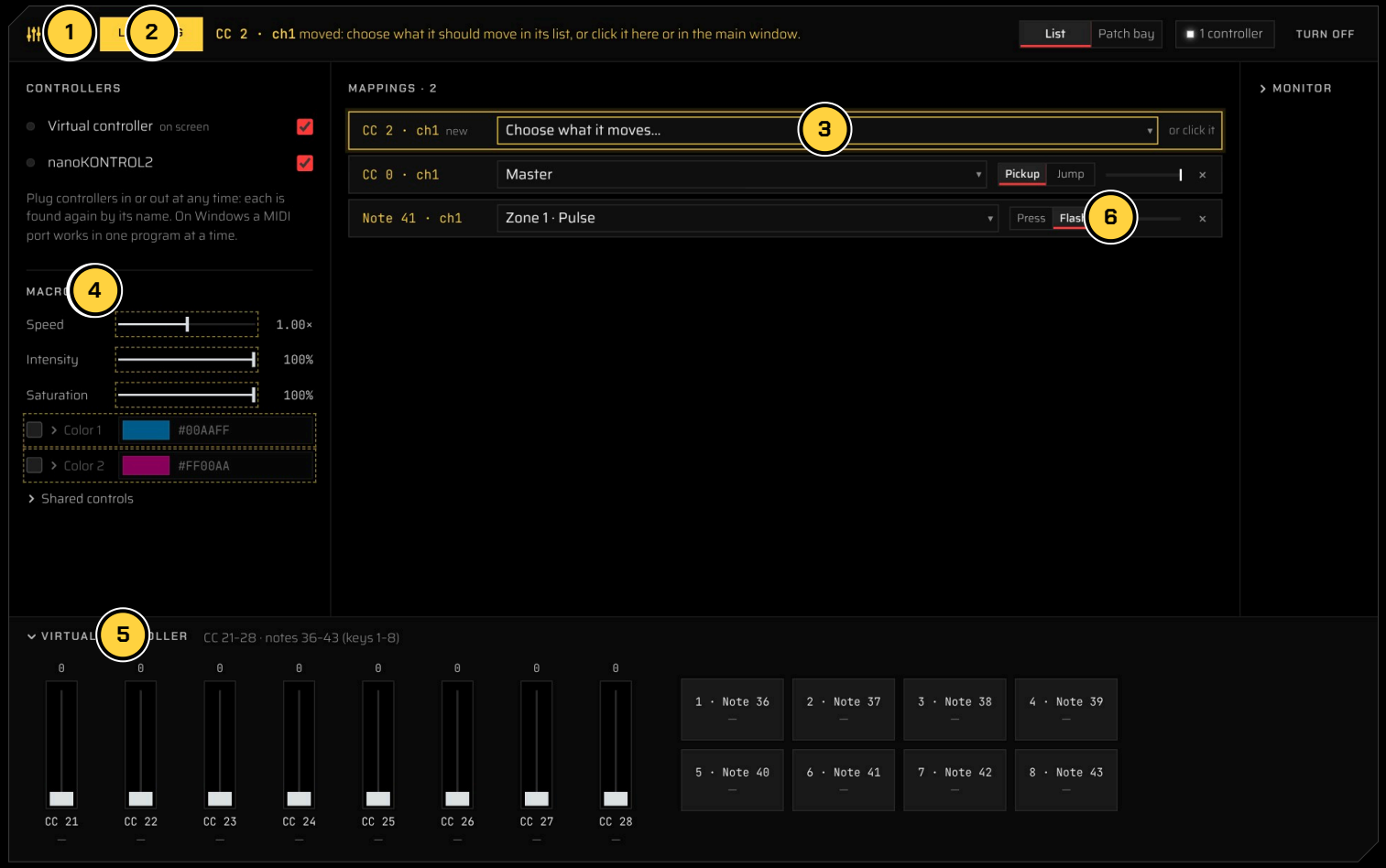
TIP Double-click a step to play from it; ⋯ → Edit playlist opens this window again. The phone remote plays playlists too.

18

MIDI

LEARN YOUR CONTROLLER

1. MIDI in the top bar opens the MIDI window, with your controllers.
2. Learn, then move a fader, knob or pad.
3. Choose what it moves from the list, or click it in nouPix. Faders pick up where the slider is.
4. Macros: Speed, Intensity, Saturation and two colours, for every effect at once.
5. Virtual controller: faders and pads on screen, to try it without hardware.
6. Flash on a pad: a scene, a level or blackout only while held.



TIP Master, blackout, the X-Fader, group dimmers, zone levels, scenes, playlists, macros and any control of an effect can be learnt.

19

PATCH BAY

EVERY MAPPING AS A WIRE

1

Patch bay: the same mappings, drawn as yellow wires.

2

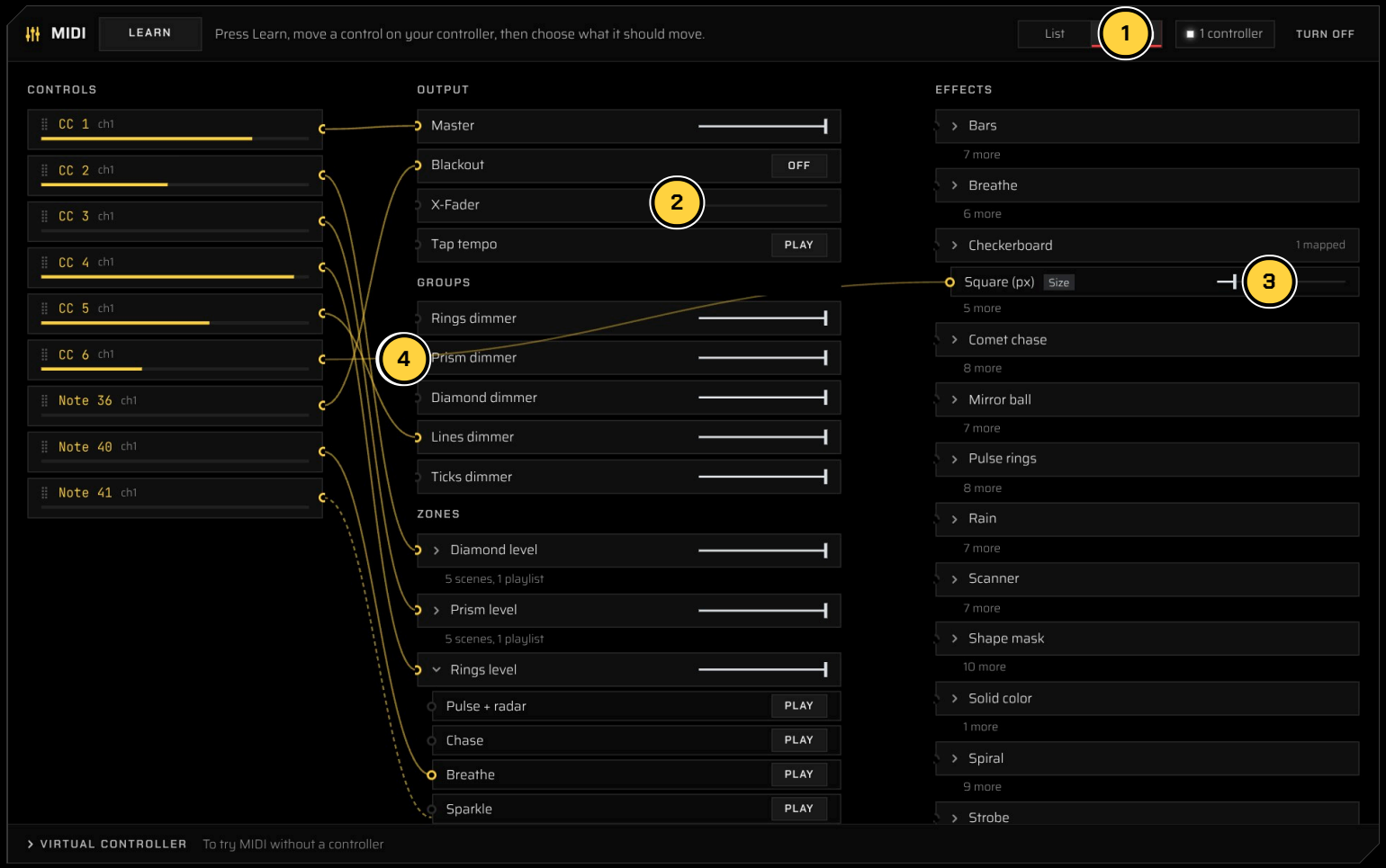
Controls on the left, what they drive in the middle: master, groups, zones, scenes, macros.

3

Effects on the right: wire any control of any effect.

4

Drag from a dot to another to wire them. Click a wire for its settings; Delete removes it.



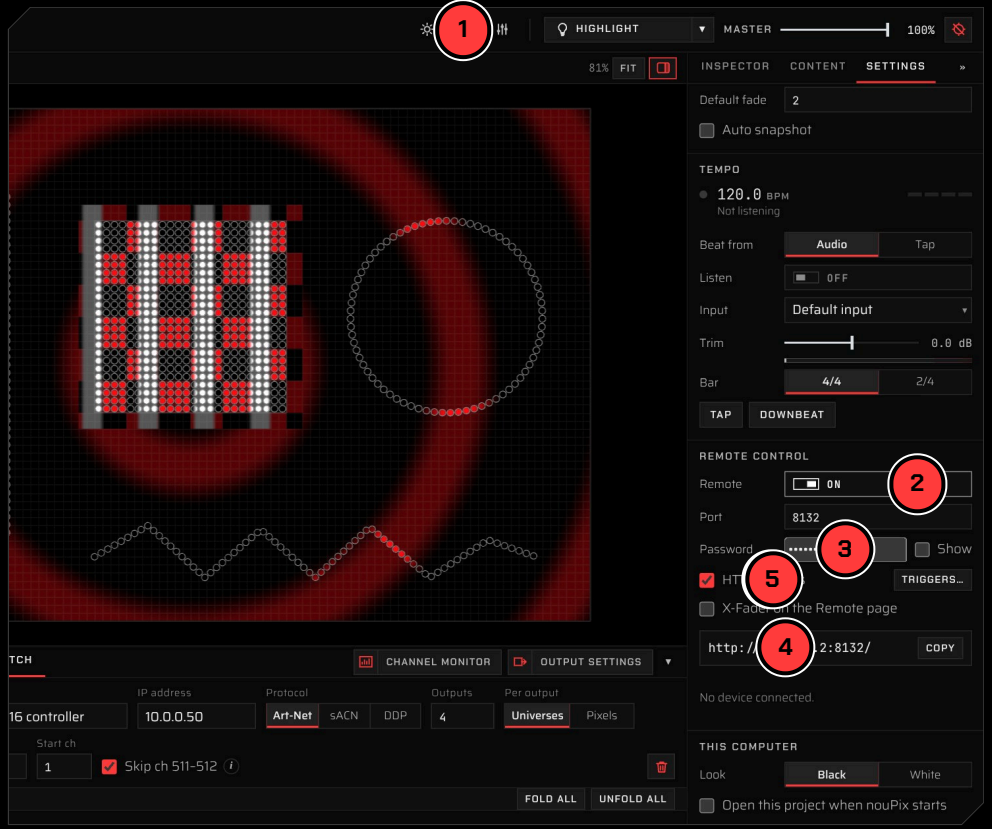
TIP Point at a macro to see, in aluminium wires, the effect controls that follow it.

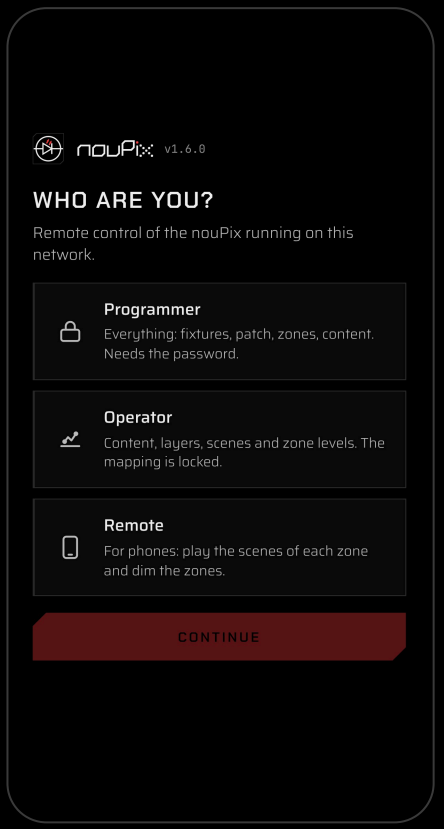
20

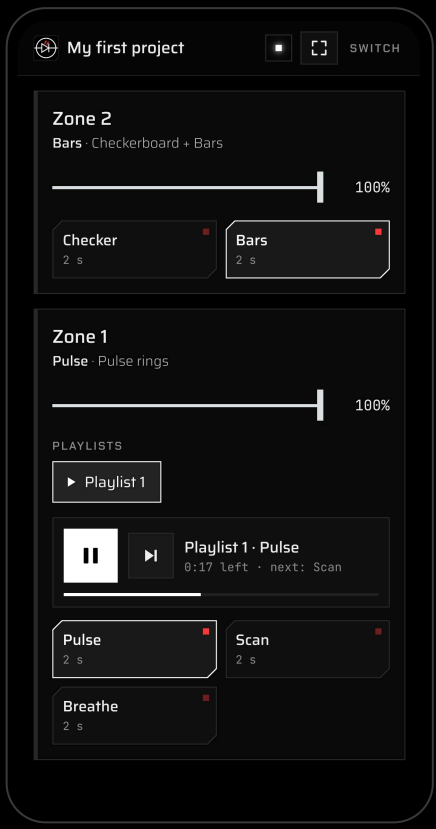
REMOTE CONTROL

PLAY IT FROM A PHONE

1. Remote in the top bar → Switch remote control on, or Settings → Remote control.
2. On: a second server opens on your network.
3. Set a password for the Programmer role.
4. Open the address on a phone on the same network. Remote plays scenes and playlists, dims zones and, if you tick it, moves the X-Fader; Operator changes content too; Programmer everything.
5. HTTP triggers: show control, a Stream Deck or a building management system plays scenes and sets levels with a web address.







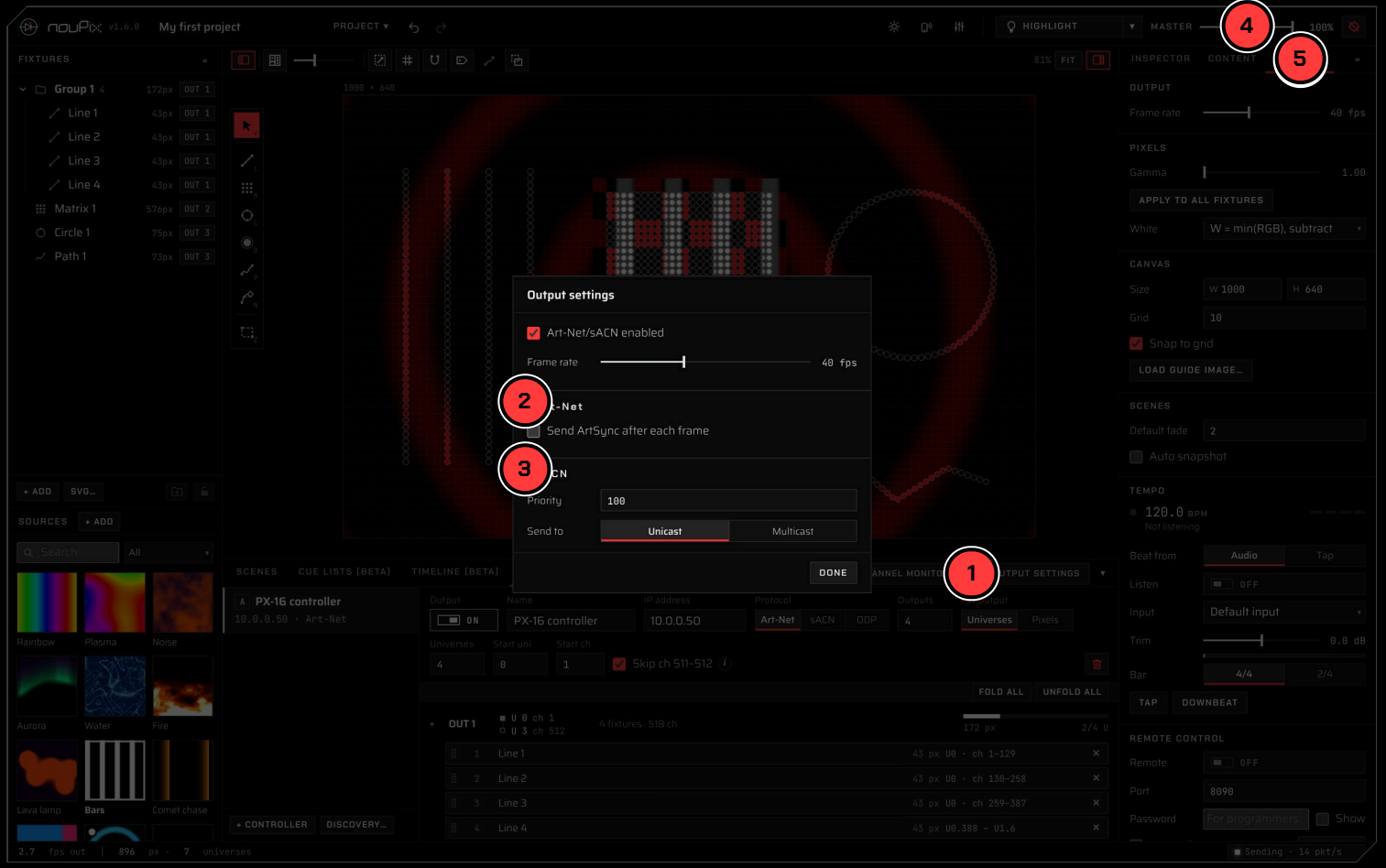
TIP With remote control on, Remote in the top bar → HTTP triggers... lists every address, ready to copy: scenes, zone levels, the master and blackout.

21

OUTPUT AND SETTINGS

MASTER, LOOK AND THE OUTPUT

1. **Output settings** (in the Patch tab): output on or off, the frame rate.
2. **Art-Net**: ArtSync after each frame, for nodes that support it.
3. **sACN**: priority, and unicast or multicast.
4. **Master** and **blackout** (B) in the top bar.
5. **Settings**: Pixels (gamma, RGBW white), Canvas, Scenes, and This computer: look, startup project, updates.



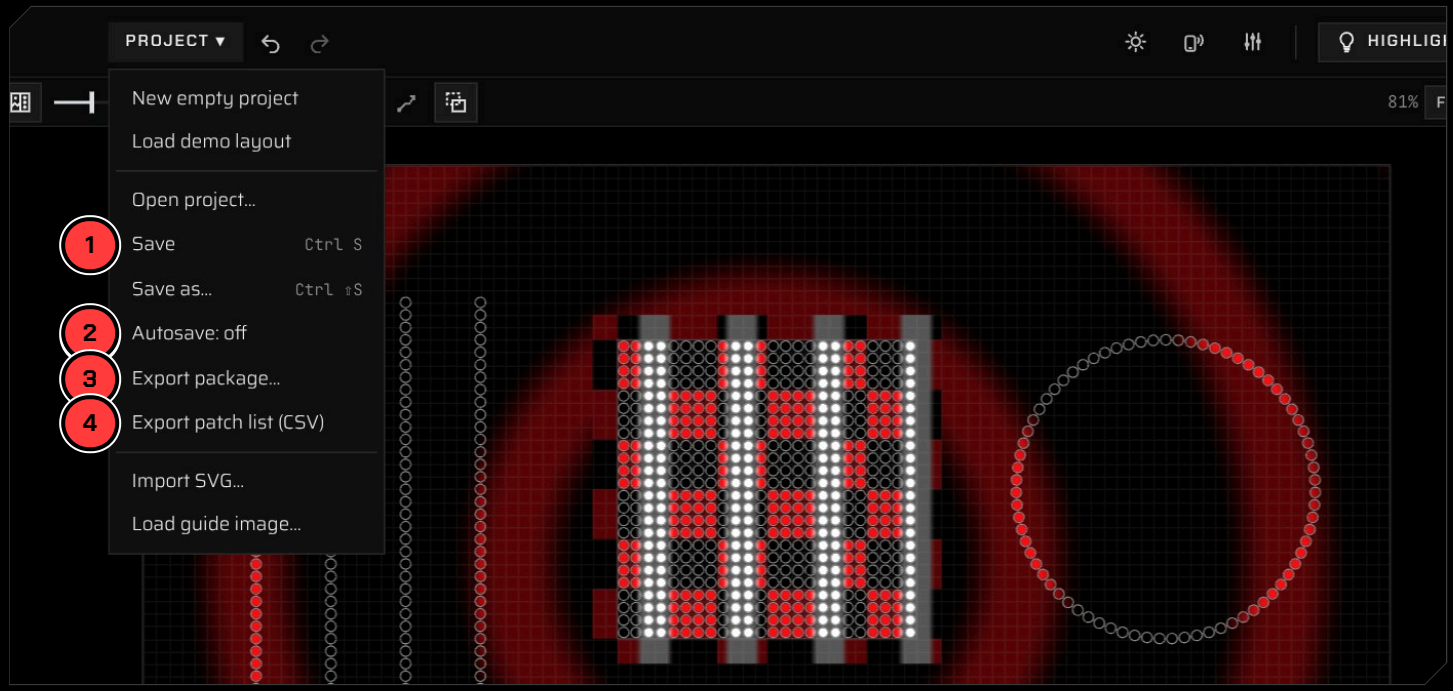
TIP Untick **Art-Net/sACN enabled** to stop the output, for example while a lighting console drives the lights. The light / dark button in the top bar switches the look.

22

SAVE AND SHARE

KEEP IT, SHARE IT

1. Save (Ctrl S) writes over the .noupix file; Save as... makes a new one.
2. Autosave writes every change to the file.
3. Export package...: one .noupixpkg file with the project and its images and videos, for another computer.
4. Export patch list (CSV): universes and addresses, to print or share.



TIP Settings → This computer → Open this project when nouPix starts, and a scene or playlist can play when it opens: a show computer runs by itself.

23

TEMPO AND THE MUSIC

FOLLOW THE BEAT

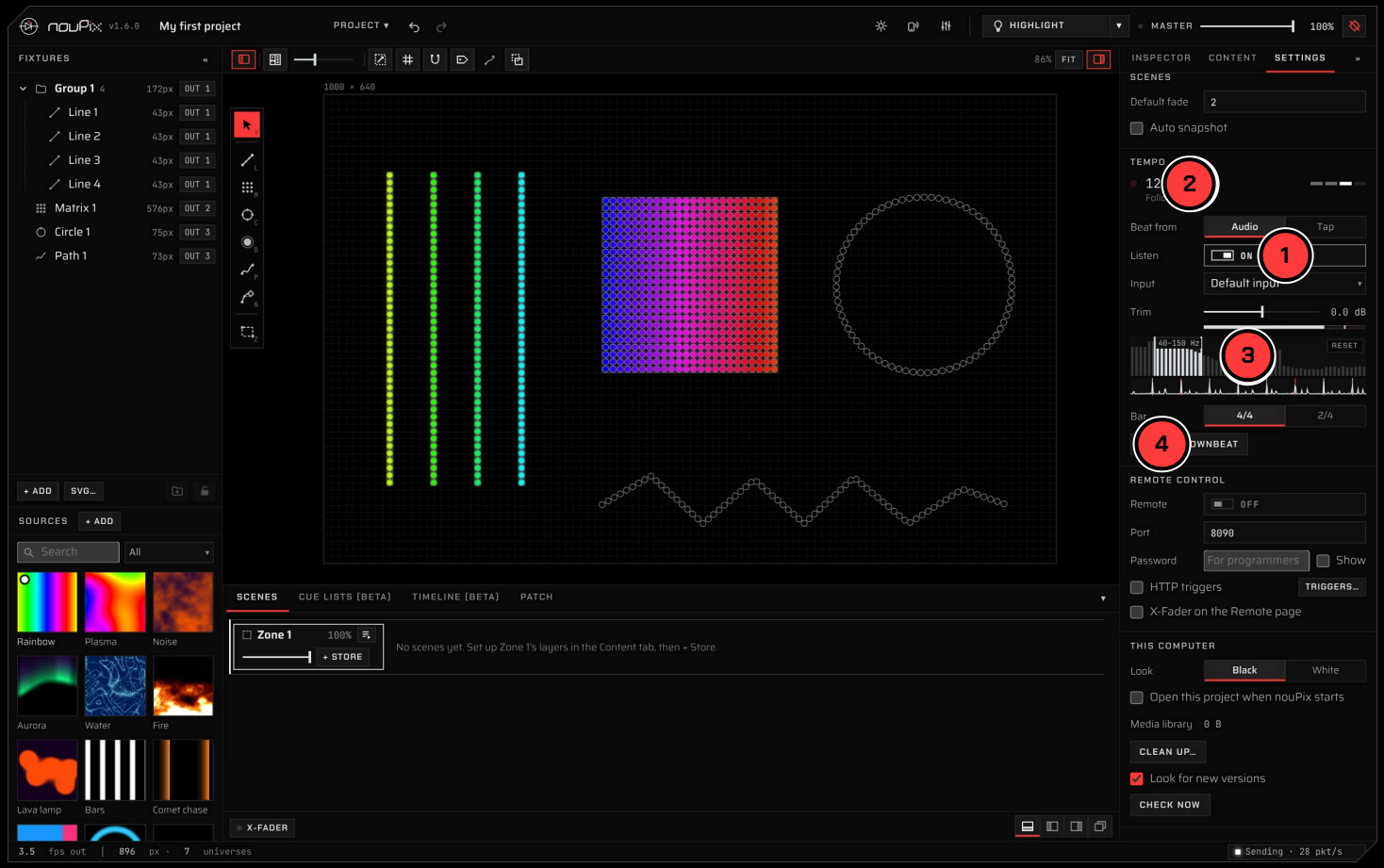
1 Settings → Tempo: switch Listen on, choose the input and set the trim.

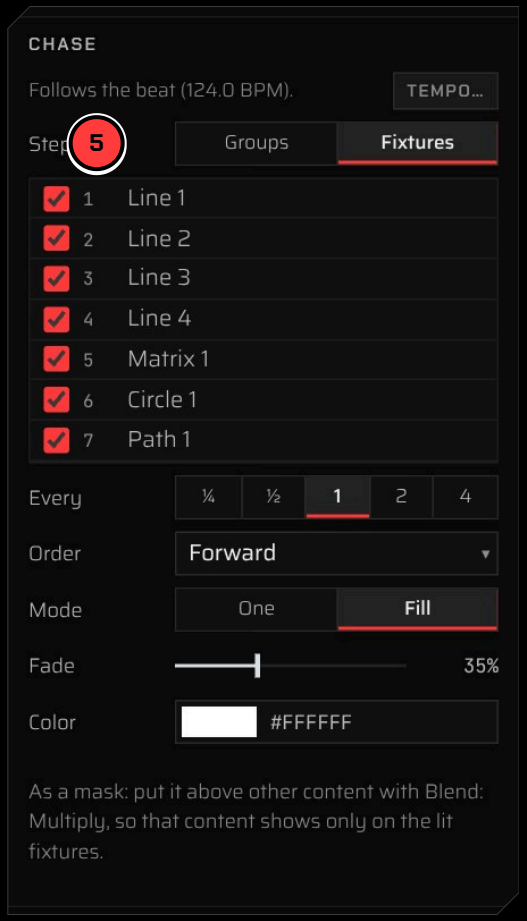
2 The BPM, with a red dot on each beat and the bar counted.

3 The beat band on the spectrum: drag it onto the kick of the music.

4 Or Tap the beat, also from a MIDI pad. Downbeat puts beat 1.

5 Fixture Chase: your groups or fixtures in turn, every ¼ to 4 beats, one at a time or filling up.





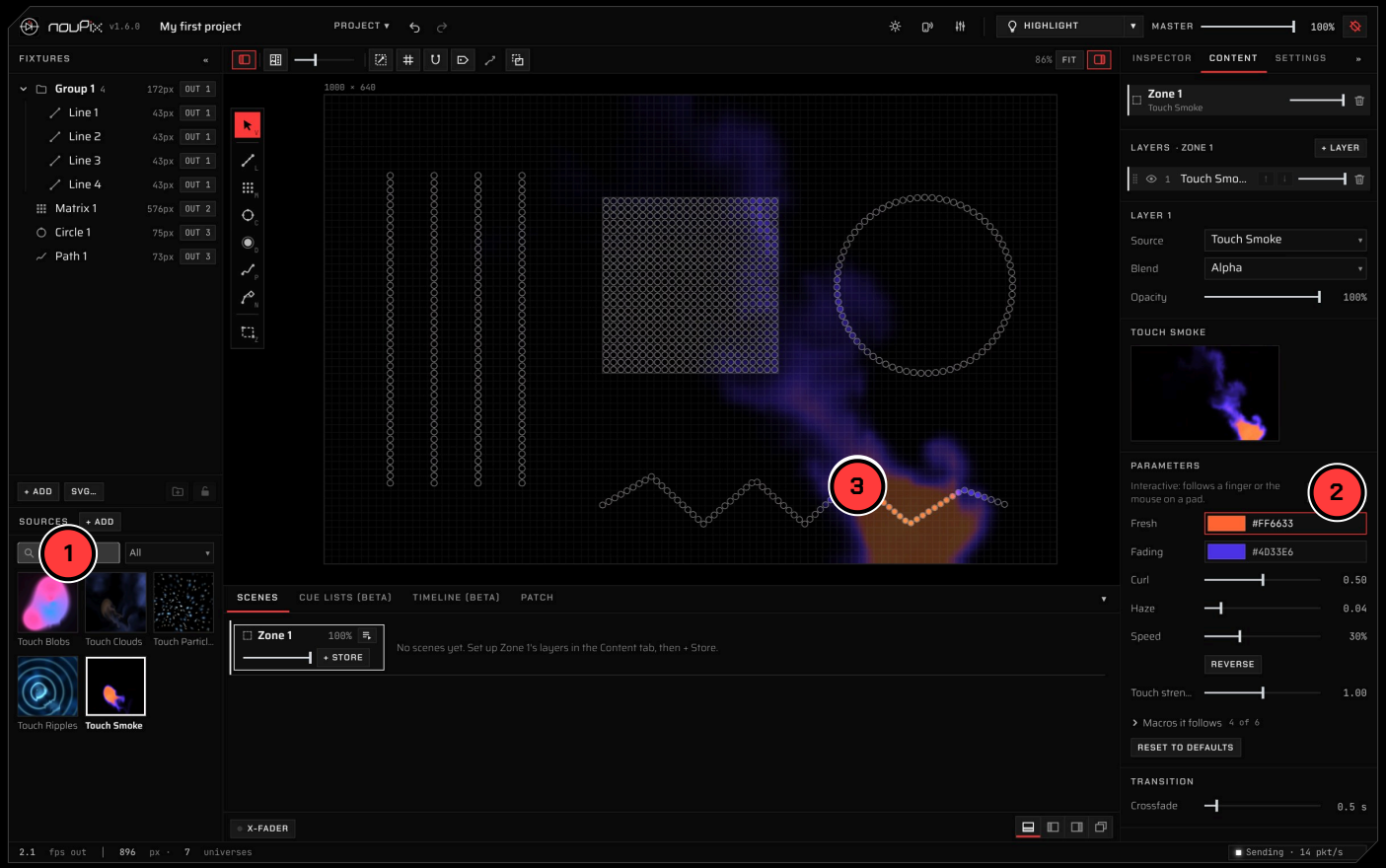
TIP Search beat in Sources for the 12 effects that follow the music. The Fixture Chase above other content with Blend: Multiply works as a mask: here it lights the rainbow fixture by fixture.

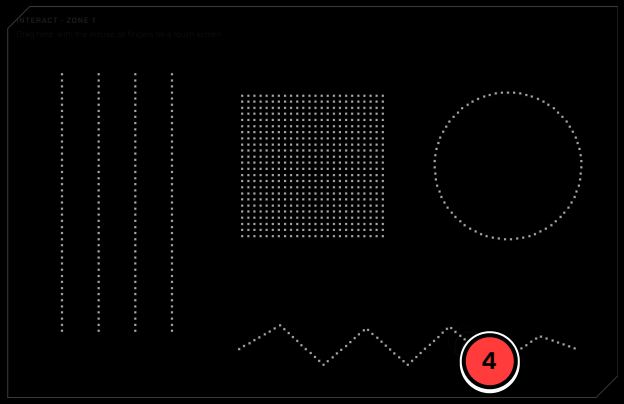
24

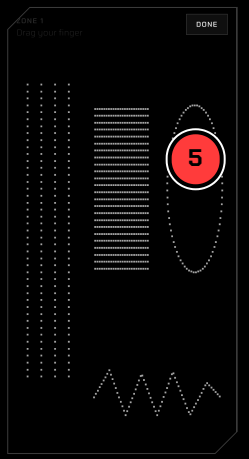
INTERACTIVE CONTENT

LET PEOPLE PLAY WITH THE LIGHT

1. Search **touch** in Sources: blobs, clouds, particles, ripples, smoke.
2. **Interact** in the effect's Parameters opens a pad in its own window.
3. **Drag on it**, with the mouse or fingers: the light follows and keeps swirling.
4. The pad shows the zone's **fixtures** faintly, so you see where to play.
5. On a phone: the **Remote** page → **Interact** makes it a full-screen pad.







TIP Untouched, the effects drift by themselves. Several pads and phones can play at once, each zone with its own content.

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CUE LISTS AND TIMELINE · BETA

CUE BY CUE, OR ON A TIMELINE

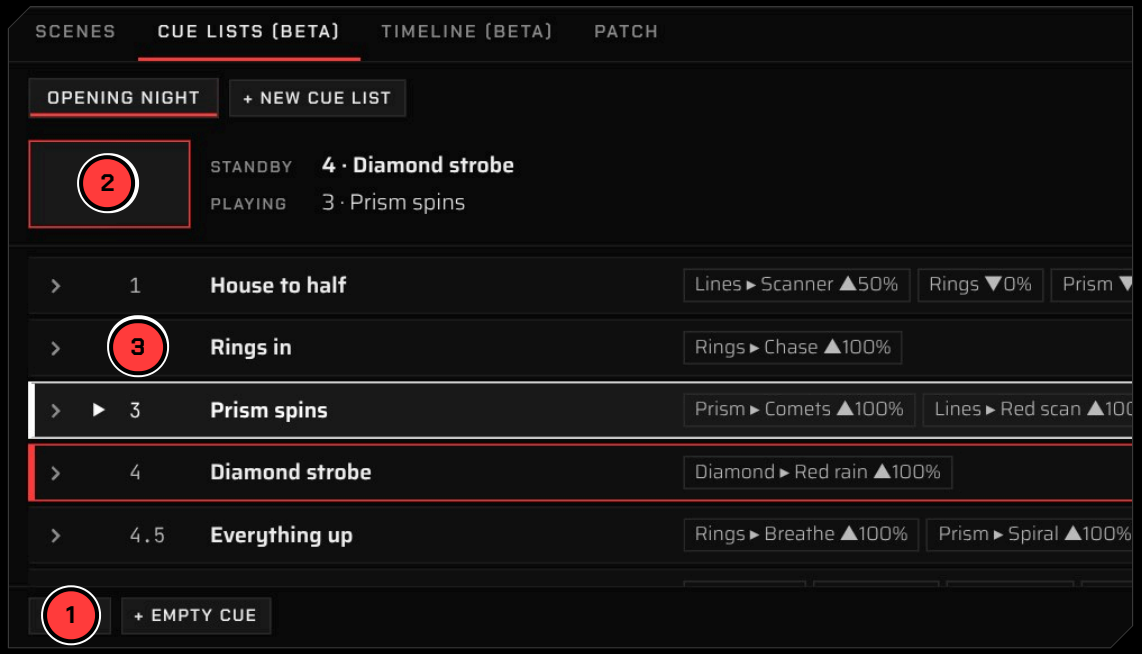
1 Cue lists tab → + Cue records what every zone shows now. Open a cue with its arrow to change its actions, fade and what comes after.

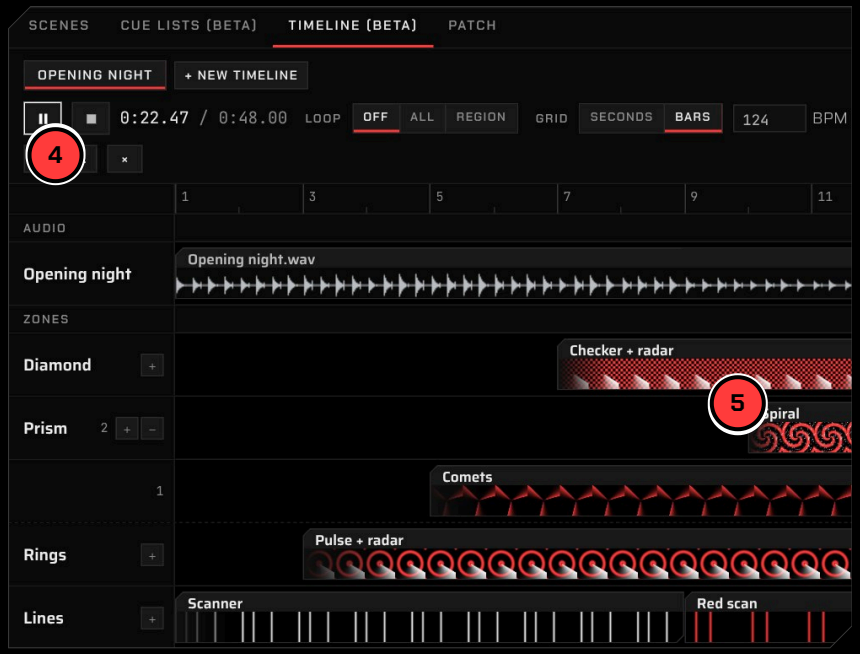
2 GO (or Space) plays the cue on standby, outlined red; the cue playing is white.

3 A cue can go on by itself after a wait, to the next or to another. Back and Stop.

4 Timeline tab: Audio… adds a sound file; drag a scene onto a zone's track as a clip.

5 Drag a clip to move it, its edges to resize, its top corners for the fades. Loop, and the ruler in bars.





TIP Both are in beta: try them, and report what you miss on GitHub. The Scenes, Cue lists and Timeline panels can move to a side panel with the buttons at their top right.

SHORTCUTS

KEYS TO REMEMBER

V	Select	L	Line	M	Matrix
C	Circle	P	Path	N	Pen
Z	Zones	Ctrl C	Copy	Ctrl X	Cut
Ctrl V	Paste	Ctrl D	Duplicate	Del	Delete
Ctrl G	Group	Ctrl Shift G	Ungroup	R	Turn 90°
Ctrl L	Lock	Ctrl Z	Undo	Ctrl S	Save
1-9	Scenes of the zone	I	Highlight	B	Blackout
F	Fit the view	H	Hide the side panel	Space	Drag to pan
D	Dot	Ctrl Shift S	Save as	Ctrl Shift Z	Redo

On a Mac, Ctrl is ⌘.

DOWNLOAD
noupix.nouled.com

SOURCE AND MANUAL
github.com/nouled-studio/nouPix

BUGS AND IDEAS
github.com/nouled-studio/nouPix/issues

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